

Hamze Hammami

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Dubai, UAE

PROFESSIONAL EXPERIENCE

- **Heriot-Watt School of Mathematics and Computer Science** Dubai, UAE
Laboratory Assistant *August 2025 – Present*
 - Provide lab support through demonstrations, tutorials and coding support
 - Laboratory assistance for Intelligent Robotics
- **Heriot-Watt School of Engineering and Physical Sciences** Dubai, UAE
Teaching Assistant *Jan 2025 – Present*
 - Provide teaching support through lab demonstrations and tutorials for robotics and programming courses
 - Teaching assistance for Machine Vision, Praxis Programming, Robotics System Sciences and Praxis Electronics
- **Vee Tech** Cairo, Egypt
Embedded Systems and ML Engineer *Jul 2023 – Dec 2023*
 - Developed company's standard intelligent systems for Automatic Number Plate Recognition and Facial Recognition Pipeline with liveness security checks
 - Collected and processed datasets for computer vision applications
 - Implemented secure embedded systems and created APIs for web and mobile integration
- **Vee Tech** Cairo, Egypt
Backend Intern *Apr 2023 – Jun 2023*
 - Received Training on NestJS Framework
 - Conducted testing of backend systems and SDKs
 - Automated backend QA testing processes

RESEARCH & PUBLICATIONS

- **SUSHI: A Vision System for Reactive, Uninformed ASV Navigation via Multi-Field Path Planning and Visual Exploration**
Journal of Intelligent & Robotic Systems
 - Status: Published
 - Authors: Hamze Hammami, Muhammad Abban, Abdul Maa'jid Aga, Laith Mohamed, Saif Alsaad, Nidhal Abdulaziz
 - DOI: 10.1007/s10846-026-02379-9
- **Why Evolve When You Can Adapt? Post-Evolution Adaptation of Genetic Memory for On-the-Fly Control**
2025 Conference on Artificial Life (ALIFE 2025)
 - Status: Presented at ALIFE 2025 and published
 - Authors: Hamze Hammami, Eva Denisa Barbulescu, Talal Shaikh, Mouayad Aldada, Muhammad Saad Munawar
 - DOI: 10.1007/978-3-031-71340-5_15
 - Track: SONI – Self-Organising Neural Intelligence(s)
- **Enhancing Creativity and Efficiency in Furniture Design through Human-AI Collaboration**
AAAI-SuS-2025 Symposium on Human-AI Collaboration (HAIC 2025)
 - Status: Presented at Heriot Watt Dubai and Published
 - Authors: Nidhal Abdulaziz, Mohamed Al Musleh, Vrushali Mhatre, Hamze Hammami
 - DOI: 10.1609/aaai.ss25.36048
- **BeeBetter: A Multi-modal Beehive System for Honeybee Health Monitoring and Hazard Detection**
2024 International Conference on Signal Processing and Information Systems (ICSPIS 2024)
 - Status: Presented at University of Dubai and Published
 - Authors: Hamze Hammami, Nidhal Abdulaziz
 - DOI: 10.1109/ICSPIS63676.2024.10812634
- **Carbide Volume Fraction in Cobalt-based Alloys using ML techniques**
International Conference on AI and Materials for Sustainability (AIMS 2025)
 - Status: Poster to be presented at IIT Delhi Abu Dhabi campus, December 2025
 - Authors: Vishakh Kumar, Hamze Hammami, Rehan Ahmed, Nidhal Abdulaziz
- **Robot Single Sound Source Localization with ITD Binaural Cue Processing Using Limited Microphone Sensors**
Preprint — TechRxiv
 - Status: Preprint
 - Contributors: Hamze Hammami, Morteza Varasteh
 - DOI: 10.36227/techrxiv.175355484.45273307/v1

EDUCATION

- **Heriot-Watt University**
MSc in Robotics Computer System Engineering 2024 – Present
- **University of Essex**
BEng (Hons.) in Computer System Engineering 2019 – 2022
- **Essex International College (Via Kaplan)**
Foundation Degree in Computer Engineering 2018 – 2019
- **Narmer American College**
American High School Diploma 2018

KEY PROJECTS

- **Bee Better/ BeeVE (Master Thesis):** Bee Better purposes a smart bee hive monitoring system with multimodal computer vision and signal processing for agricultural automation applications, BEEve is an extension related to the signal processing part which proposes a A Vector-Quantized Encoder for Unsupervised Non-Semantic State Discovery through representation learning which became the focus of the Thesis. | **Tech: Python, non-human communication, pattern discovery, decoding acoustics, representation learning, latent space, Computer Vision, Signal Processing**
- **AISV – Autonomous Surface Vessel:** Team Lead for developing autonomous software for sea debris collection vessel featuring novel seabin-inspired mechanism. Integrated Vision Fusion system, Hybrid Path Planning and Exploration, and Behavior Control using Fuzzy Logic and Dynamic Window approach into ROS framework. Received 12K AED funding through Mubadala competition. | **Tech: ROS2, SkFuzz, Python, TensorRT, Jetson Orin, RViz**
- **Hebbian Robot Lifetime Control:** Researched two-phase learning approach for robot control through MLP neural networks using Webots simulation. E-puck robots learned navigation tasks in empty T-shaped environments, then adapted to environmental changes by applying neuroscience principles that mimic human brain learning for on-the-fly reactions to new obstacles. | **Tech: Webots, Hebbian Learning, Evolutionary Programming, MLP, E-puck**
- **Eye Tracking Camera:** Designed and implemented 2-axis camera mechanism for real-time facial tracking with precise servoing through 360° servos on embedded hardware. | **Tech: Inventor (AutoCAD), Google Coral, TensorFlow Lite**
- **Facial Recognition Pipeline:** Implemented facial recognition database using Redis for user verification with liveness detection, deployed across private residence security systems. | **Tech: Python, Redis, Computer Vision**
- **Automatic Number Plate Recognition (ANPR) System:** Built multi-detection system for extracting vehicle color and license plate information through YOLO detection and plate number extraction. | **Tech: Python, YOLO, TensorFlow**
- **Sound Localization (Bachelor Capstone Project):** Developed robot audition system for sound azimuth localization using binaural sound processing with two-microphone array. | **Tech: Arduino, EMoRo Robot, Binaural Sound Processing**

TALKS & VOLUNTEERING

- **Earth Species Project Community Deep Dive — BeeVE Acoustic State Discovery** Online
Invited Speaker May 2026
 - Invited by the Earth Species Project (ESP) Community to present BeeVE, covering unsupervised analysis of honey bee buzzing.
 - Recording: Zoom link
- **Why Evolve When You Can Adapt? Post-Evolution Adaptation of Genetic Memory** Kyoto, Japan / Online
Presenter — ALIFE 2025, SONI Track Jul 2025
 - Presented at the 2025 Conference on Artificial Life (ALIFE 2025), track: Self-Organising Neural Intelligence(s). In-person presentation held in Kyoto; pre-recorded talk delivered via online.
 - Recording: Pre-Recorded presentation
- **Enhancing Creativity and Efficiency in Furniture Design through Human-AI Collaboration** Heriot-Watt Dubai, UAE
Presenter — AAAI-SuS-2025, HAIC Symposium May 2025
 - Co-authored on this project but was Presented by me at the AAAI 2025 Symposium on Human-AI Collaboration (HAIC 2025); co-author on the published paper.
 - Recording: Google drive self recorded
- **The Second YOLO Workshop (Space Theme)** UoBD, Dubai, UAE
Invited Presenter and Demonstrator Mar 2025
 - Invited by Emirates Students Space Association (ESSA), University of Birmingham Dubai Space Society, and UoBD ACM Chapter to demonstrate Ultralytics YOLO framework applications in space technology
- **The First YOLO Workshop** Heriot-Watt, Dubai, UAE
Event Presenter and Demonstrator 2025
 - Taught students YOLO framework implementation, AI and computer vision concepts through hands-on activities and practical ML library demonstrations

- **Demonstration on Path Planning** Heriot-Watt, Dubai, UAE
Invited Demonstrator 2025
 - Delivered technical Demonstration on path planning and motion planning algorithms for robot arms and differential drive robots to mechanical and robotics engineering students
- **Heriot-Watt "Engineer for a Day"** Heriot-Watt University, Dubai
Volunteer 2024 -
 - Led Arduino programming and obstacle-detection workshops for high-school visitors during university outreach events
 - Operating the University's Uni-Tree GO1 dog robot for or high-school visitors in multiple events
- **IROS 2024 Conference Volunteer** Abu Dhabi, UAE
Technical Volunteer Oct 2024
 - Supported hardware-focused demonstrations and interactive robotics competitions at IROS 2024 (Khalifa University)

SKILLS

- **Programming Languages:** C++, Python, C, MATLAB, C#, Arduino, Node.js
- **Embedded AI Toolchain:** CUDA, TensorRT, JetPack 6, DeepStream, Jetson Orin/TX2, Raspberry Pi
- **AI & Machine Learning:** Computer Vision, Neural Networks, Fuzzy Logic, Image & Signal Processing, YOLO, PyTorch, TensorFlow
- **Robotics & Automation:** ROS2, Path Planning (D* Lite), SLAM, MoveIt, Single-board computers, IoT systems
- **Development Tools:** $\LaTeX$, Conda, Linux, RViz, Redis, Git
- **Leadership & Management:** Project research, team leadership, technical mentoring, cross-functional collaboration

CERTIFICATIONS & TRAINING

- **Deep Reinforcement Learning:** ROS Developers Academy – Certificate of Excellence, Jul 2024. **Verified Certificate**
- **ROS Winter Camp:** Dubai Future Labs & The Construct – 5-day intensive on MoveIt, SLAM, mapping, and OpenCV in ROS2, Feb 2024
- **Embedded Machine Learning:** Edge Impulse – Certificate Earned, May 2022

LANGUAGES

- **Arabic:** Native Language
- **English:** Advanced Proficiency